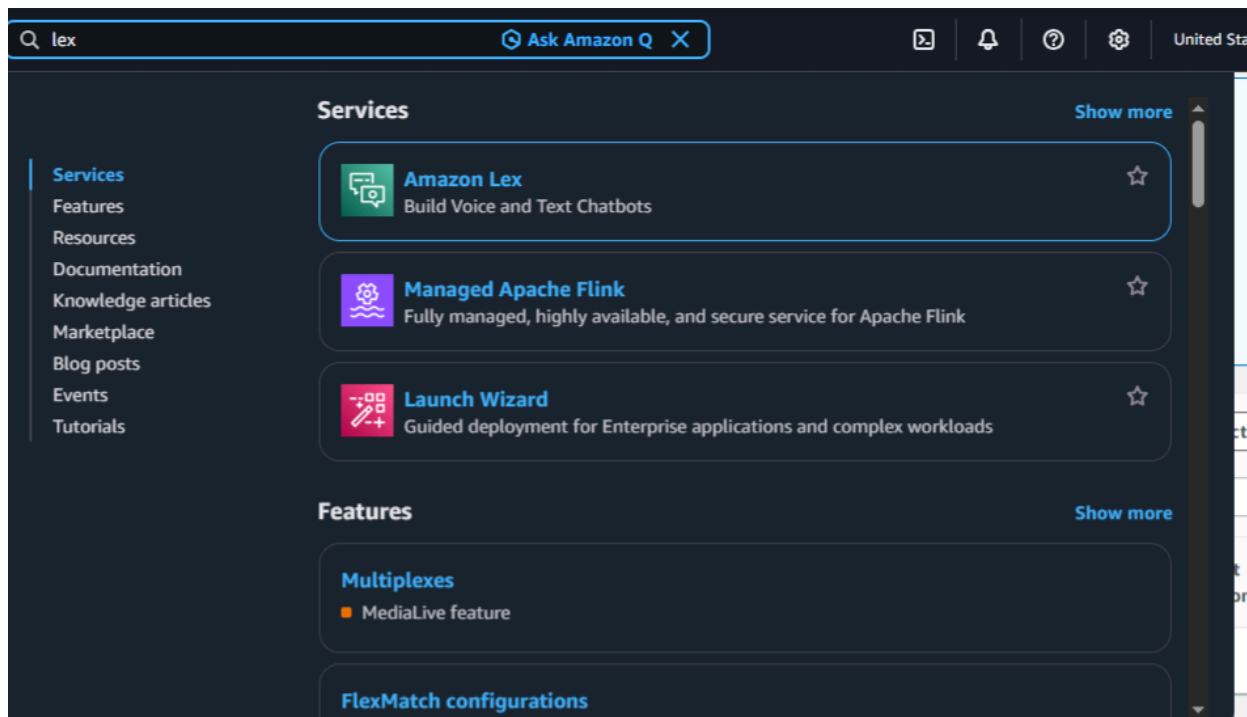


AWS S3 Quiz Chatbot with Amazon Lex

I built an interactive knowledge-check chatbot for Cloud Learners Inc. using Amazon Lex V2. The bot quizzes users on Amazon S3 with multiple-choice questions and provides instant feedback. Users can click answer buttons instead of typing, making the experience engaging and intuitive.

Step-by-Step Implementation

I signed in to the AWS Management Console, searched and opened Amazon Lex.



I clicked Create bot, then chose "Create a blank bot". I named my bot **Lex_QuizBot**

Bots (1) Info

Action ▼

Create bot

Q Search bots

< 1 > ⚙

Name ▲	Description	Status ▼	Latest Version ▼	Last updated ▼
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☒ **Create a blank bot**
Create a basic bot with no preconfigured languages, intents, and slot types.

☐ **Start with an example**
An example bot has preconfigured languages, intents, and slot types. You can change these settings.

☐ **Start with transcripts**
Automatically generate intents from conversation transcripts that you upload. Only English (US) language is available when starting with a transcript.

Bot configuration

Bot name

Lex_QuizBot

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Description - optional

This description appears on bot list page. It can help you identify the purpose of your bot.

IT HelpDesk bot for employees in the North America office.

Maximum 2000 characters.

Left the following in their default settings:

IAM permissions: "Create a role with basic Amazon Lex permissions"

COPPA: "No"

Then I clicked Next

Children's Online Privacy Protection Act (COPPA) [Info](#)

Is use of your bot subject to the [Children's Online Privacy Protection Act \(COPPA\)](#) [?](#)

☐ Yes

☒ No

Idle session timeout

You can configure how long a session is maintained when the user does not provide any input and the session is idle. Amazon Lex retains context information until a session ends.

Session timeout

By default, session duration is 5 minutes, but you can specify any duration between 1 and 1440 minutes (24 hours).

► **Advanced settings - optional** [Info](#)

Cancel

Next

Selected English (US) as the language, and clicked Done

Step 1

Configure bot settings

Step 2

Add languages

Add language to bot [Info](#)

▼ Language: English (US)

Select language

English (US)

Description - optional

Maximum 2000 characters.

☐ Enable speech-to-speech inference

Voice interaction

The text-to-speech voice that your bot uses to interact with users.

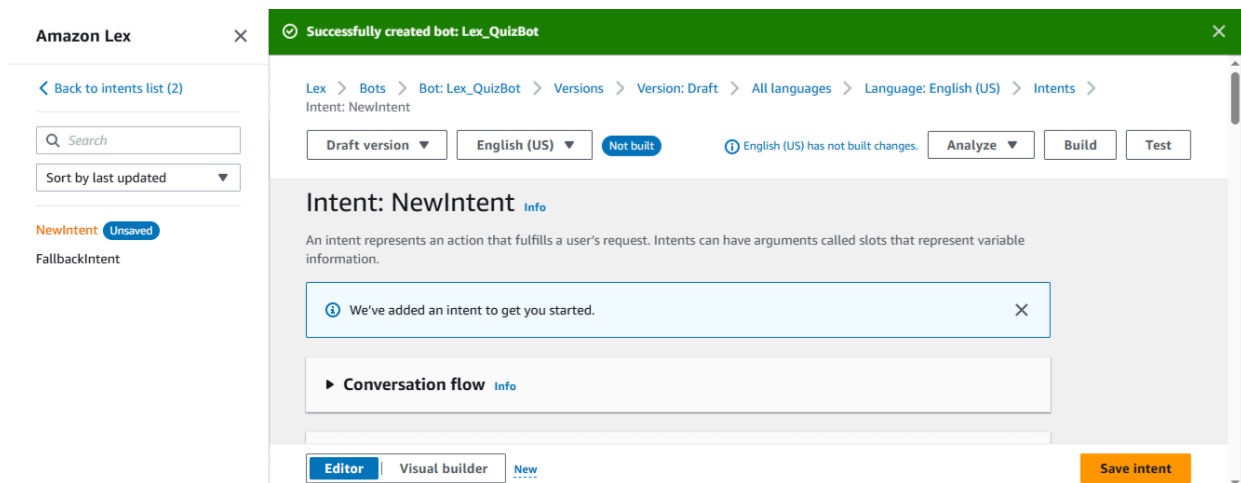
Danielle

Cancel

Add another language

Done

Once my Bot was created, I skipped **Conversation Flow** and navigated to **Intent Details**



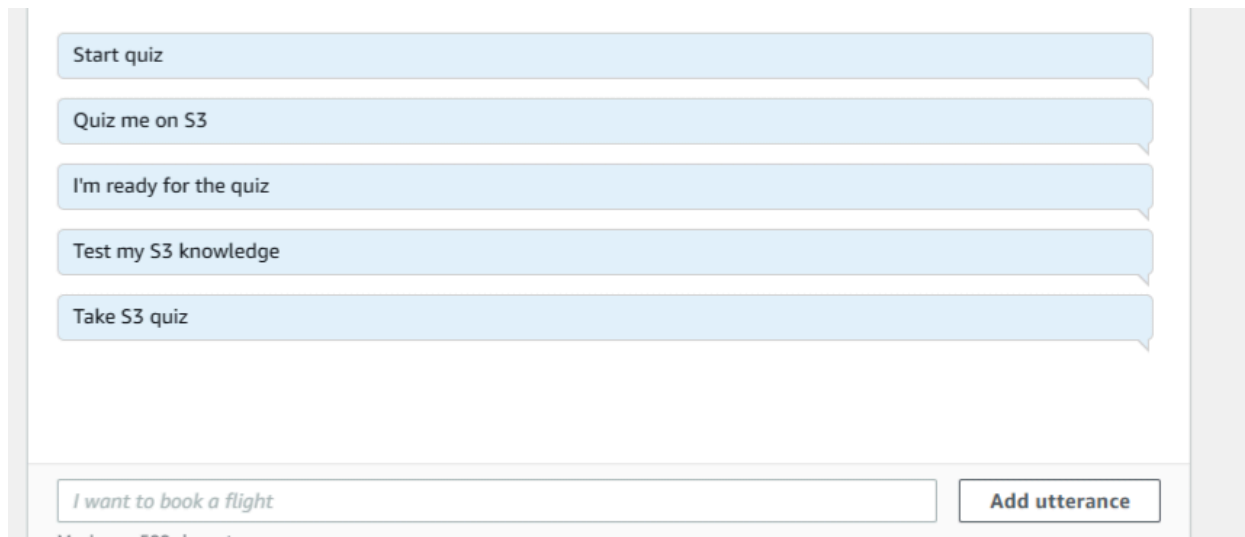
To create my intent I named it : S3 Quiz and in the gave the same title for the **Display Name**.

The screenshot shows the 'Intent details' form. The section is titled 'Intent details' with an 'Info' link. It contains two input fields. The first is 'Intent name' with the value 'S3Quiz' and a note: 'Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _'. The second is 'Display name' with the value 'S3Quiz' and a note: 'Recommended for Intent Disambiguation. Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _'.

Navigated down to Sample Utterances, where i entered the following phrases to trigger the Quiz:

- "Start quiz"
- "Quiz me on S3"
- "I'm ready for the quiz"
- "Take S3 quiz"

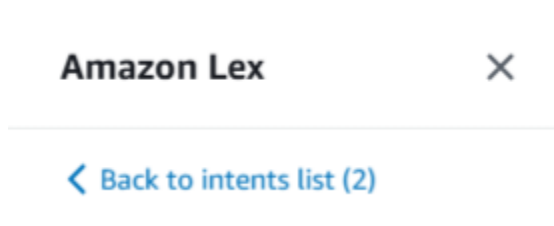
- "Test my S3 knowledge"



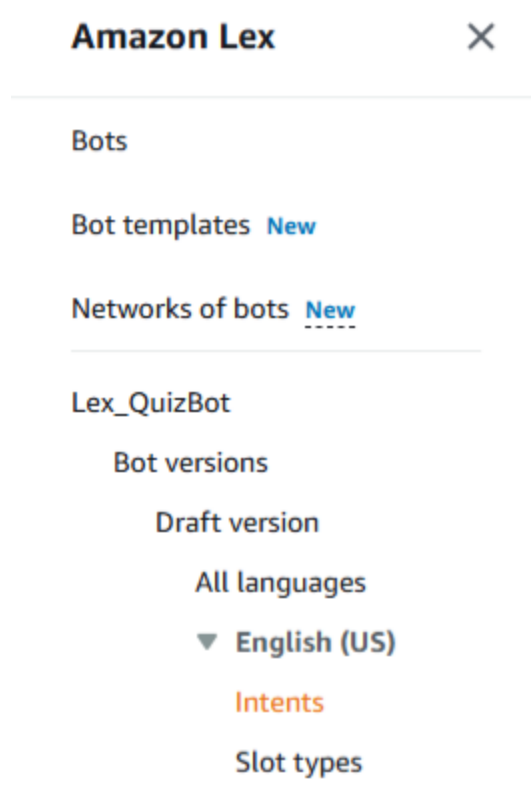
The screenshot shows a list of five intents in a light blue box. The intents are: "Start quiz", "Quiz me on S3", "I'm ready for the quiz", "Test my S3 knowledge", and "Take S3 quiz". Below the list is a text input field containing the text "I want to book a flight" and a button labeled "Add utterance".

Scrolled down and saved my intent.

On the left pane clicked on **Back to intents list**

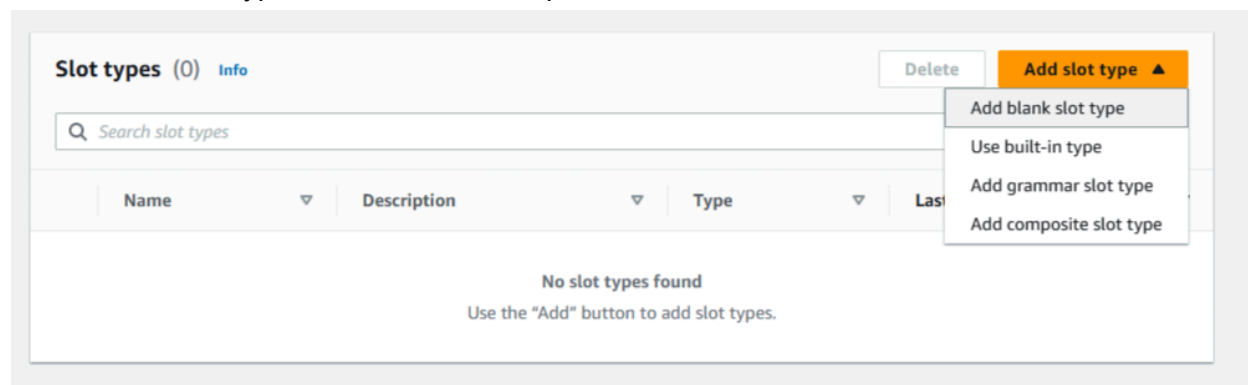


I went back to the pane and clicked **Slot types**.



I created a custom slot type to define the possible answer choices for the quiz questions. This restricts user input to valid options (A, B, or C) and allows me to use synonyms so Lex can understand variations like "Simple Storage Service" as equivalent to "A".

I clicked Add slot type, named it S3QuizOptions



Add blank slot type

×

Create a custom slot type for your bot.

Slot type name

S3QuizOptions

Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _

Cancel

Add

For **Slot value resolution** I selected **Restrict to slot values**, which restricted the user to only pick from the predefined options.

Under slot type values i entered my answer options for the first question of the quiz

Value: A → : Simple Storage Service

Value: B → : Secure Server Storage

Value: C → : Smart Storage System

Slot type values

Modify the list of values used to train the machine learning model to recognize values for a slot.

Q Search slot type values

A

Tab or ; or enter return for new value

×

Simple Storage Service

×

B

Tab or ; or enter return for new value

×

Secure Server Storage

×

C

Tab or ; or enter return for new value

×

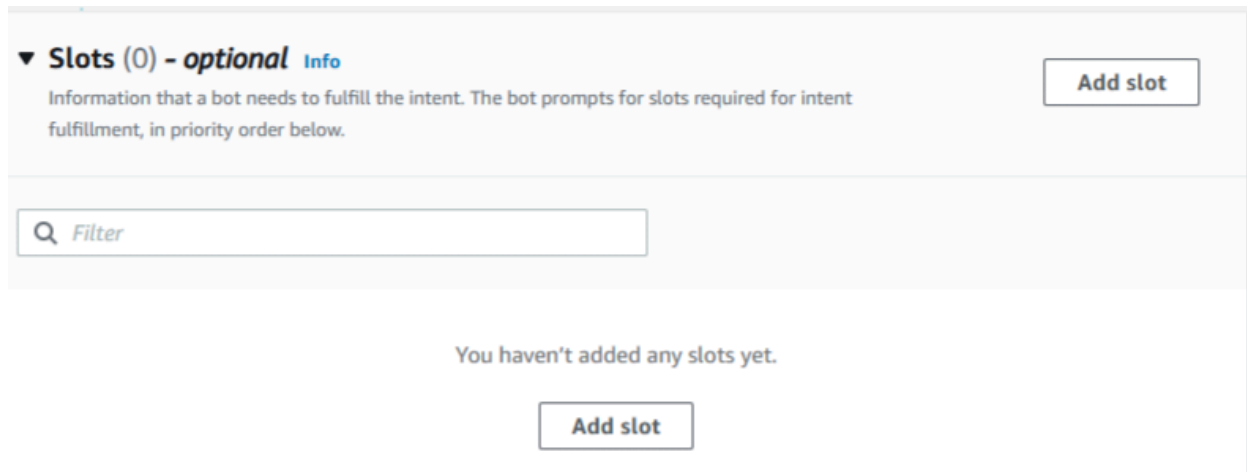
Smart Storage System

×

Then saved slot type.

Clicked on **Slot Types** on the left pane and navigated back to my intent

After clicking back on my intent i scrolled down to **Slots**



▼ **Slots (0) - optional** [Info](#)

Information that a bot needs to fulfill the intent. The bot prompts for slots required for intent fulfillment, in priority order below.

[Add slot](#)

Q Filter

You haven't added any slots yet.

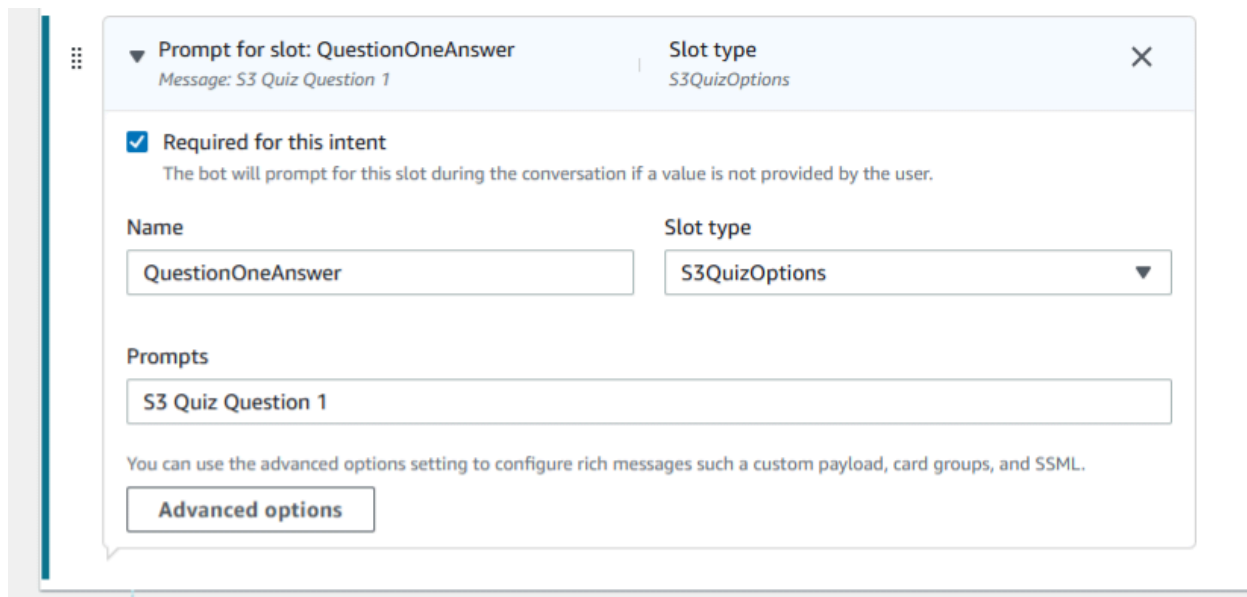
[Add slot](#)

I clicked **Add slot** and named it QuestionOneAnswer

For **Slot Type** i selected the one i created "S3QuizOptions"

In the **Prompt** section. I entered the prompt message: S3 Quiz Question 1

Then I Clicked on **Advanced Options**



⋮

▼ Prompt for slot: QuestionOneAnswer
Message: S3 Quiz Question 1

Slot type
S3QuizOptions

☒ **Required for this intent**
The bot will prompt for this slot during the conversation if a value is not provided by the user.

Name
QuestionOneAnswer

Slot type
S3QuizOptions ▼

Prompts
S3 Quiz Question 1

You can use the advanced options setting to configure rich messages such a custom payload, card groups, and SSML.

[Advanced options](#)

Scrolled down to **Slot Prompts** clicked on the Prompt and selected **More prompt options**. Navigated down to **Slot Prompts** again and clicked on **Add**, selected **Add Card Group**.

Slot: QuestionOneAnswer

Info

✕

▼ Bot elicits information

Message: S3 Quiz Question 1

☐ Play the messages in order

Info

Messages will be used in the predefined order as slot prompts by your bot.

▼ Message group

Info

You can define a text message group to respond using plain text.

Message

S3 Quiz Question 1

► Variations - optional

More prompt options

1 more message group: 1 card group.

Slot: QuestionOneAnswer

> Slot prompts editor

✕

Slot prompts

Prompts to elicit the slot.

☐ Preview

Add ▲

Add text message group

Add card group

Add custom payload

Add SSML language

▼ Message group

Info

✕

You can define a text message group to respond using plain text.

Message

S3 Quiz Question 1

► Variations - optional

Under **Card Group**, I filled in:

Title: What Does S3 Stand For?

Subtitle: Please Select A, B or C

For Button 1, A) Simple Storage Service and filled in A as the value. Did the same for B and C

Slot: QuestionOneAnswer > Slot prompts editor

▼ Card group Info

You can define up to three cards per group. One of the cards is selected during a conversation.

Card 1 + Add card

You can use response card with Facebook Messenger, Slack, Twilio, or your own client applications.

Image URL

Must be an Amazon S3 object URL.

Title

Maximum 250 characters. Valid characters: A-Z, a-z, 0-9, @, #, \$

Subtitle

Slot: QuestionOneAnswer > Slot prompts editor

▼ Buttons - optional

Button 1 title	Button 1 value
A) Simple Storage Service	A
Button title can have up to 50 characters.	Button value can have up to 50 characters.
Button 2 title	Button 2 value
B) Secure Server Storage	B
Button title can have up to 50 characters.	Button value can have up to 50 characters.
Button 3 title	Button 3 value
C) Smart Storage System	C
Button title can have up to 50 characters.	Button value can have up to 50 characters.

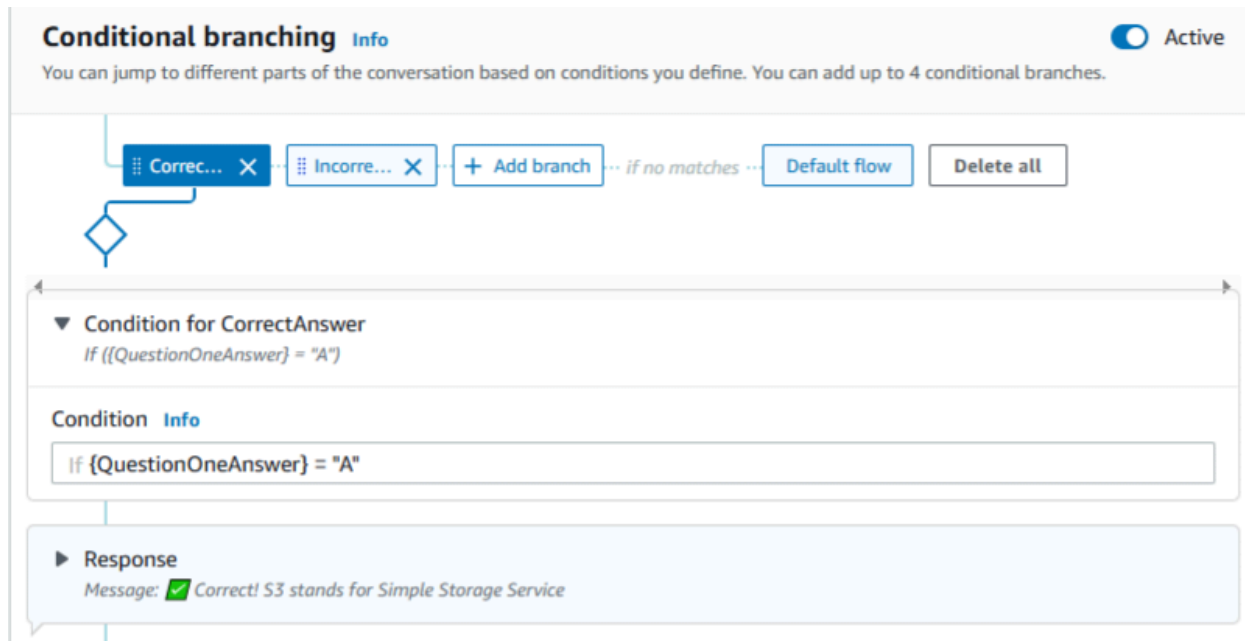
After filling in the response card, I clicked the **Add** button at the bottom of the configuration panel to save the prompt settings. And further clicked **Save Slot**

I scrolled to the **Closing response** section and toggled **Active** to ON.

Navigated down to **Conditional Branching** renamed **branch 1** to **Correct Answer**

And under **Condition** typed **{QuestionOneAnswer} = "A"**

For **Response** typed in **Correct! S3 stands for Simple Storage Service**



Clicked on add branch and renamed it **Incorrect Answer**

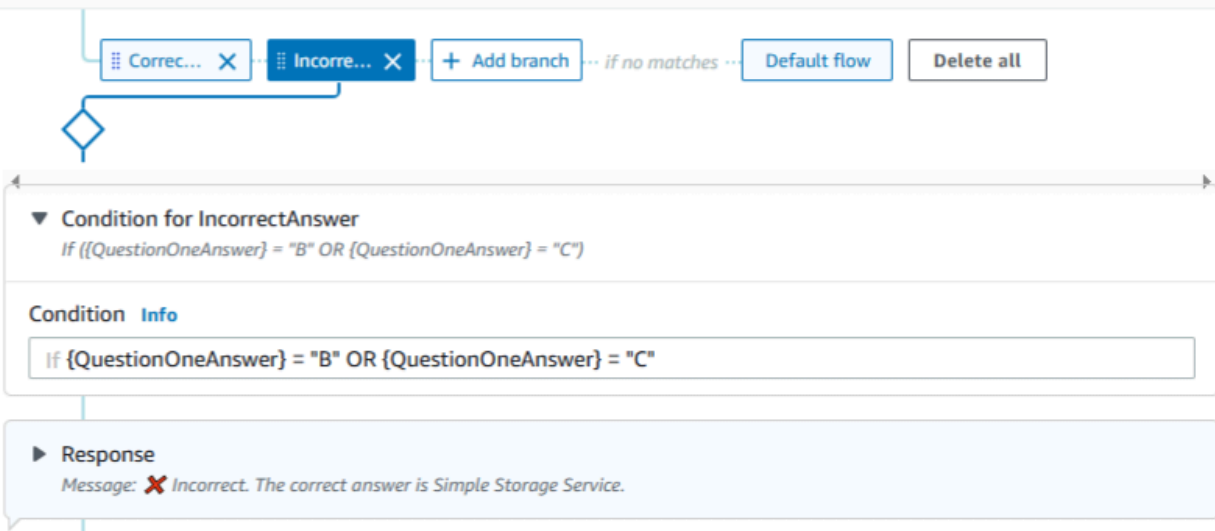
Under **Condition** typed **{QuestionOneAnswer} = "B"**

And for **Response** typed in **Incorrect. The correct answer is Simple Storage Service.**

Conditional branching [Info](#)

☒ Active

You can jump to different parts of the conversation based on conditions you define. You can add up to 4 conditional branches.



Beta Testing

Inspect

Start quiz

S3 Quiz Question 1

What Does S3 Stand For?

Please Select A, B or C?

**A) Simple
Storage Service**

✓ Ready for complete testing



Type a message

Inspect

Storage Service

B) Secure Server
Storage

C) Smart
Storage System

A

✓ Correct! S3 stands for Simple
Storage Service

✓ Ready for complete testing



Type a message

Inspect

Storage Service

B) Secure Server
Storage

C) Smart
Storage System

B

✗ Incorrect. The correct answer
is Simple Storage Service.

✓ Ready for complete testing



Type a message